C++ Midway Project: Student Record Manager

Ahmed Mohammed

What the program does

The Student Record Manager keeps track of students and their math scores. Anyone can view the students, search by id or name, see class statistics, and write a report to a file. An Admin can also add students, update scores, and delete students. The Admin password is admin123 for this demo.

How it works

The Roster class loads students.csv into a dynamic array when the program starts. The array stays sorted by id so binary search can find a student by id. Search by name goes through the array one student at a time. If the array fills up, it is replaced with one twice as big. Each add, update and delete saves back to students.csv.

Files

main.cpp (login, menu and the menu actions), Roster.h / Roster.cpp (the array, searches, statistics and file I/O), Student.h / Student.cpp (one student), User.h (the User, Admin and RegularUser classes), students.csv (the data).

Concepts used

Data types and sizes	int, double, char, bool and string. Menu option 5 prints their sizes with sizeof.
File types	students.csv holds the data and report.txt is a text report.
Pointers	Student* for the dynamic array, findById() returns a pointer, getAverage() moves a pointer through the array, and the menu functions take a Roster*.
Dynamic memory	new[] in the Roster constructor and grow(), delete[] in grow() and the destructor.
Arrays	The students array, plus the letters and gradeCounts arrays in showStats().
Search	Roster::binarySearch() finds a student by id. searchByName() is a linear search.
Strings	Names, getline, and find() for searching part of a name.
File I/O	loadFromFile() reads with ifstream. saveToFile() and writeReport() write with ofstream.
Classes	Student, Roster, User, Admin and RegularUser. Admin and RegularUser inherit from User.

Screenshots of the program running

1. Starting up and logging in as Admin

```
Loaded 8 students.
Enter your name: Ahmed
1. Admin
2. Regular User
Choose your role: 1
Enter admin password: admin123
```

2. Main menu

```
=== STUDENT RECORD MANAGER ===
Logged in as Ahmed (Admin)
1. View Students
2. Search by ID
3. Search by Name
4. Class Statistics
5. Data Types and Sizes
6. Write Report to File
7. Add Student (Admin)
8. Update Score (Admin)
9. Delete Student (Admin)
0. Exit
Choose an option (0-9):
```

3. View Students: the list is sorted by id

```
Choose an option (0-9): 1
Ada (id 1001) - 72, grade C
Sam (id 1002) - 69, grade D
Jordan (id 1003) - 90, grade A
Riley (id 1004) - 47, grade F
Casey (id 1005) - 76, grade C
Morgan (id 1006) - 71, grade C
Avery (id 1007) - 88, grade B
Quinn (id 1008) - 40, grade F
```

4. Search by ID: found

```
Choose an option (0–9): 2
Enter student id: 1006
Morgan (id 1006) – 71, grade C
```

5. Search by ID: not found

```
Choose an option (0–9): 2
Enter student id: 2500
No student with that id.
```

6. Search by Name: found

```
Choose an option (0–9): 3
Enter a name: Ri
Riley (id 1004) – 47, grade F
```

7. Search by Name: not found

```
Choose an option (0–9): 3
Enter a name: Zoe
No student with that name.
```

8. Class Statistics

```
Choose an option (0–9): 4
Students: 8
Average: 69.125
Highest: Jordan – 90
Lowest: Quinn – 40
Passing: 6
A: 1
B: 1
C: 3
D: 1
F: 2
```

9. Data Types and Sizes

```
Choose an option (0–9): 5
char: 1 byte
bool: 1 byte
int: 4 bytes
double: 8 bytes
string: 24 bytes
Student: 40 bytes
Student pointer: 8 bytes
The array has room for 10 students, which is 400 bytes.
```

10. Write Report to File

```
Choose an option (0–9): 6
Report written to report.txt.
```

11. The report.txt file the program wrote

```
$ cat report.txt
Student Report
Students: 8
Ada (id 1001) – 72, grade C
Sam (id 1002) – 69, grade D
Jordan (id 1003) – 90, grade A
Riley (id 1004) – 47, grade F
Casey (id 1005) – 76, grade C
Morgan (id 1006) – 71, grade C
Avery (id 1007) – 88, grade B
Quinn (id 1008) – 40, grade F
Average: 69.125
```

12. Add Student

```
Choose an option (0–9): 7
Enter student name: Taylor Reed
Enter student id: 1009
Enter math score (0–100): 84.5
Student added.
Saved to students.csv.
```

13. Add Student: the id is already used

```
Choose an option (0–9): 7
Enter student name: Duplicate
Enter student id: 1003
That id is already used.
```

14. Add Student: the score is out of range

```
Choose an option (0–9): 7
Enter student name: Bad Score
Enter student id: 1010
Enter math score (0–100): 150
The score has to be from 0 to 100.
```

15. Update Score

```
Choose an option (0–9): 8
Enter student id: 1004
Enter math score (0–100): 63
Riley (id 1004) – 63, grade D
Saved to students.csv.
```

16. Delete Student

```
Choose an option (0–9): 9
Enter student id: 1008
Student deleted.
Saved to students.csv.
```

17. View Students after the changes

```
Choose an option (0–9): 1
Ada (id 1001) – 72, grade C
Sam (id 1002) – 69, grade D
Jordan (id 1003) – 90, grade A
Riley (id 1004) – 63, grade D
Casey (id 1005) – 76, grade C
Morgan (id 1006) – 71, grade C
Avery (id 1007) – 88, grade B
Taylor Reed (id 1009) – 84.5, grade B
```

18. The students.csv file after the changes

```
$ cat students.csv
name,student_id,math_score
Ada,1001,72
Sam,1002,69
Jordan,1003,90
Riley,1004,63
Casey,1005,76
Morgan,1006,71
Avery,1007,88
Taylor Reed,1009,84.5
```

19. Invalid menu input

```
Choose an option (0–9): abc
Invalid choice. Pick 0–9.

Choose an option (0–9): 42
Invalid choice. Pick 0–9.
```

20. Exit

```
Choose an option (0–9): 0
Goodbye!
```

21. Wrong admin password: logged in as a Regular User

```
Loaded 8 students.
Enter your name: Kim
1. Admin
2. Regular User
Choose your role: 1
Enter admin password: oops
Wrong password. Logging in as Regular User.
```

22. Logging in as a Regular User

```
Loaded 8 students.
Enter your name: Sam
1. Admin
2. Regular User
Choose your role: 2

=== STUDENT RECORD MANAGER ===
Logged in as Sam (Regular User)
1. View Students
2. Search by ID
3. Search by Name
4. Class Statistics
5. Data Types and Sizes
6. Write Report to File
7. Add Student (Admin)
8. Update Score (Admin)
9. Delete Student (Admin)
0. Exit
Choose an option (0-9):
```

23. A Regular User can view and search

```
Choose an option (0–9): 1
Ada (id 1001) – 72, grade C
Sam (id 1002) – 69, grade D
Jordan (id 1003) – 90, grade A
Riley (id 1004) – 63, grade D
Casey (id 1005) – 76, grade C
Morgan (id 1006) – 71, grade C
Avery (id 1007) – 88, grade B
Taylor Reed (id 1009) – 84.5, grade B

Choose an option (0–9): 2
Enter student id: 1002
Sam (id 1002) – 69, grade D
```

24. A Regular User is denied options 7, 8 and 9

```
Choose an option (0–9): 7
Access denied. Only an Admin can do that.

Choose an option (0–9): 8
Access denied. Only an Admin can do that.

Choose an option (0–9): 9
Access denied. Only an Admin can do that.

Choose an option (0–9): 0
Goodbye!
```